

Verified compiler for a language with control effects

(Weryfikacja kompilatora dla języka programowania z efektami sterowania)

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Abstract

We present a type system for a language with labeled delimited control operators. It utilizes polymorphic types and effects to allow expressions parametrized by effects. Our goal is to formally prove correctness of such language. Formalisation work was done in Agda programming language.

Tematem tej pracy jest stworzenie i zweryfikowanie języka z operatorami sterowania dla kontynuacji ograniczonych. Używa on polimorficznych typów aby pozwolić na wyrażenie obliczeń generycznych wobec pewnych efektów. Celem pracy jest zweryfikowanie poprawności definicji takiego języka. Formalizacja została przeprowadzona w języku programowania Agda.

Contents

1	Introduction	7
2	Surface language	9
2.1	Kinds, Types, Effects and Expressions	9
2.2	Contexts	10
2.3	Typing judgements	10
3	Runtime language	13
3.1	Runtime language	13
3.2	Runtime embedding	14
4	Progress	17
4.1	Values	17
4.2	Reduction	19
4.3	Progress	20
4.3.1	Preservation	21
5	Discussion/Closing thoughts	23
	Bibliography	25

Chapter 1

Introduction

Control operators have been present in languages for a long time. One notable example is SCHEME operator programming language which started in 70s[1] and provides *call/cc* operator. They enable non-local jumps, that might be familiar to users of imperative languages with *goto* operator. But those operators unlike *goto*, are handled by capturing continuation, and storing it as first-class value which can be passed, stored or dropped. One issue one might have with *call/cc* operator is that when called, it catches whole rest of program in the continuation. For example in `(call/cc ( $\lambda k \rightarrow k$  ( $k$  1)))` only inner k would be called.

Operators for delimited control introduced by Danvy, and Filinski[2] and independently by Felleisen[3] allow to control continuations them more precisely. They usually come in pairs, for example operator *reset* delimits continuations caught by *shift*. This allows for localized use of control effects, scope of captured continuation is much more visible. While they still allow for non-local execution, which enables interesting computational effects such as emulating non-determinism, exceptions, failure and others. Usually such calculi jump from *shift* to closest enclosing *reset*. Reduction rule looks like this $(\text{reset } E[(\text{shift } k \text{ expr})]) \Rightarrow (\text{reset } ((\lambda k \rightarrow \text{expr}) (\lambda v \rightarrow (\text{reset } E[v]))))$, so for example `(reset (+ 1 (shift k (k (k 0)))))` would call k twice and would result in value 2.

In their article on delimited continuations Danvy and Filinski[4] also describe similar operators *shift_0* and *reset_0*. Their reduction doesn't introduce outermost *reset_0*, so subsequent calls to *shift_0* remove innermost *reset_0* one by one. They used example `reset_0(3 + reset_0(4 * shift_0 k in shift_0 c in E))` to showcase this behaviour. Continuation k would be bound to $\lambda x \rightarrow 4 * x$ and c to $\lambda x \rightarrow 3 + x$. Dybvig et al.[5] have shown formalization of delimited control with multiple prompts - where operators are matched with labels.

In this thesis we try to formalize typed version of such calculus with polymorphic types (similar to Piróg, Polesiuk and Sieczkowski[6]), but for labeled delimited continuations (like Kazuki Ikemori, Youyou Cong, and Hidehiko Masuhara[7]).

Main difference to Ikemori et al. is that labels are first class like in Xie et al.[8] though for delimited continuations not for algebraic effects. Such language was present in unpublished notes of Balik, and Polesiuk[9] and currently serves as core of Fram programming language. Formalising such calculus could serve as step in proving soundness of Fram's implementation.

Soundness of presented language is proved using standard technique of progress and preservation[10][11]. Formalisation work was done in Agda 2.8. Code archive has also nix derivation describing exact environment to build both package and this paper to ensure reproducibility.

Chapter 2

Surface language

2.1 Kinds, Types, Effects and Expressions

Variable and type variables are de Bruijn indices.

```
data Kind : Set where
  T : Kind
  E : Kind
```

Types and Effects are defined mutually recursive. Type is either variable, arrow, forall or label. Effect is list of types, it's used in arrow and forall constructors of type to keep effects of computation underneath and it will be used in typing judgement. Label constructor just stores type variable bound to effect represented by the label and then Type and Effect of delimited computation.

```
module Types where
  Id : Set
  Id = ℕ
  data Type : Set
  Effects : Set
  data Type where
    ttv : Id → Type
    -->_ : Type → Effects → Type → Type
    forallt : Kind → Type → Effects → Type
    Lat_/_ : Type → Type → Effects → Type
  Effects = List Type
open Types
```

Constructors of expressions such as **var**, **lam**, **app**, **tlam**, **tapp** behave as usual. **var** is de Bruijn indexed variable, **lam** is lambda abstraction, **app** is function application, **tlam** is type abstraction, but it stores Kind of abstracted type and **tapp** is type application. The rest of constructors are responsible for continuations and labels. Constructor **new** bind label that is used by **shift** and **reset** constructors to pair up. **shift₀** stores label and expression (parametrized by continuation) that replaces whole delimited computation. Last constructor of expressions is **reset₀** which has 3 fields, last one is label that is used to match up reset with shifts. First argument is delimited computation where **shift₀** can be used. So when **shift₀ k.e** is being evaluated it aborts enclosing corresponding reset, instead of it now


```

→ Δ, Γ ⊢ app e1 e2 :: B / E

⊢forall : ∀ {Γ Δ e k A E F}
→ (Δ, k), Γ ⊢ e :: TypeSubst.bump A / TypeSubst.bump' E
-----
→ Δ, Γ ⊢ tlam k e :: forallt k A E / F

⊢tapp : ∀ {Γ Δ e k A T E}
→ Δ ⊢ T :: t
→ Δ, Γ ⊢ e :: forallt k A E / E
-----
→ Δ, Γ ⊢ tapp e T :: A TypeSubst.[ T ] / (E TypeSubst.offs[t T ])

```

new introduces new type variable, and variable that represent respectively effect and label. Type of label stores effect bound by label (here **ttv zero**). $A1 / A2$ represent type and effect of delimited computation.

```

⊢new : ∀ {Γ Δ e A A1 E E1}
→ (Δ, Kind.E), (Γ, (L ttv zero at A1 / E1)) ⊢ e :: TypeSubst.bump A / TypeSubst.bump' E
-----
→ Δ, Γ ⊢ new e :: A / E

```

shift₀ uses only one effect **ttv n** that's represented by label. For it to be properly typed expression inside shift bind extra variable - where continuation will be plugged into. So type of this expression should take continuation and returns same type as reset, with effects visible in reset. And continuation itself should be an arrow from type of shift to type of whole delimited computation with effects of same computation. Since continuation passed there will have **reset₀**, that **reset₀** will introduce effect **ttv n**, so it shouldn't be represented in type of argument.

```

⊢shift0 : ∀ {Γ Δ e e' A A' n E'}
→ Δ ⊢ ttv n :: e
→ Δ, Γ ⊢ e' :: (L ttv n at A' / E') / nil
→ Δ, (Γ, A - E' > A') ⊢ e :: A' / E'
-----
→ Δ, Γ ⊢ shift0 e' e :: A / (ttv n :: nil)

```

reset₀ has three parameters, first is expression that will have access to effect, so its list of effects is expanded. Second is continuation that will handle value returned from first argument. And third is label, which type stores type and effects of whole expression.

```

⊢reset0 : ∀ {Γ Δ e e' en A A' n E'}
→ Δ ⊢ ttv n :: e
→ Δ, Γ ⊢ e :: A / (ttv n :: E')
→ Δ, (Γ, A) ⊢ en :: A' / E'
→ Δ, Γ ⊢ e' :: (L ttv n at A' / E') / nil
-----
→ Δ, Γ ⊢ reset0 e en e' :: A' / E'

```


Chapter 3

Runtime language

3.1 Runtime language

Since grammar of terms doesn't have any expression that would have type of label, and shift and reset require same labels, that would mean that reduction relation would need to go under **new** binders. Instead we will define another expression language expanded by notion of label values.

When **new** expression is evaluated then all occurrences of variables bound by it would have it replaced with newly allocated label value. That means evaluation would need to keep a state for allocator.

To ensure type safety we will expand syntax of types with allocated effects, and add new effect context to typing judgments, that maps allocations to type and effect of delimiter.

```
module Runtime where

module Types_ where
  Id : Set
  Id = N
  Label = N
  data Type : Set
  Effects : Set
  data Type where
    ttv : Id → Type
    -->_ : Type → Effects → Type → Type
    forallt : Kind → Type → Effects → Type
    L_at_/_ : Type → Type → Effects → Type
    Effect : Label → Type -- allocated effect
    Effects = List Type
  open Types_
```

Most of constructors in **RExpr** are the same as in **Expr**. Labels runtime values are represented by Natural numbers.

```
module RuntimeExpr where
  data RExpr : Set where --runtime version
    var : N → RExpr
    lam : RExpr → RExpr
    app : RExpr → RExpr → RExpr
    tlam : Kind → RExpr → RExpr
```

We defined embedding of original language in current one, and proof that such embedding preserves typing judgements. Part of proof is omitted as it goes through almost all judgement types.

```

module Transform where
  open Types.Typing
  open Types.Expr
  open RuntimeExpr
  open Typing
  rt-t : Types.Types.Type → Type --runtime-type
  rt-t' : Types.Types.Effects → Effects
  rt-t (Types.Types.ttv x) = ttv x
  rt-t (x Types.Types.- x₁ > x₂) = rt-t x - rt-t' x₁ > rt-t x₂
  rt-t (Types.Types.forallt k x ef) = forallt k (rt-t x) (rt-t' ef)
  rt-t (Types.Types.L l at x / ef) = L rt-t l at rt-t x / rt-t' ef
  rt-t' nil = nil
  rt-t' (x :: xs) = rt-t x :: rt-t' xs
  runtime : Types.Expr.Expr → RExpr
  runtime (var x) = var x
  runtime (lam x) = lam (runtime x)
  runtime (app x x₁) = app (runtime x) (runtime x₁)
  runtime (tlam x x₁) = tlam x (runtime x₁)
  runtime (tapp x x₁) = tapp (runtime x) (rt-t x₁)
  runtime (new x) = new (runtime x)
  runtime (shift₀ x x₁) = shift₀ (runtime x) (runtime x₁)
  runtime (reset₀ x x₁ x₂) = reset₀ (runtime x) (runtime x₁) (runtime x₂)

  runtimeΔ : Types.Typing.TContext → Typing.TContext
  runtimeΔ ∅ = ∅
  runtimeΔ (Δ , k) = runtimeΔ Δ , k
  runtimeΓ : Types.Typing.Context → Typing.Context
  runtimeΓ ∅ = ∅
  runtimeΓ (Γ , T) = runtimeΓ Γ , rt-t T

runtime-types : ∀ {Δ Γ e T E}
  → Δ Types.Typing. , Γ ⊢ e :: T / E → (runtimeΔ Δ ; ∅' ; runtimeΓ Γ ⊢ (runtime e) :: rt-t T / rt-t' E)

runtime-types (⊢-var x) = ⊢-var ( runtime x )
runtime-types (⊢-lam t) = ⊢-lam (runtime-types t)
runtime-types (⊢-weak x x₁ t) = ⊢-weak (runtime<t> x) (runtime<x> x₁) (runtime-types t)
runtime-types (⊢-app t t₁) = ⊢-app (runtime-types t) (runtime-types t₁)
runtime-types (⊢-forall t) = ⊢-forall (rt-bump (runtime-types t))
runtime-types (⊢-tapp {A = A} x t) = rt-tapp {A = A} (⊢-tapp (runtime-t x) (runtime-types t))
runtime-types (⊢-new t) = ⊢-new ( rt-bump( runtime-types t) )
runtime-types (⊢-shift₀ x t t₁) = ⊢-shift₀ ((runtime-e x)) (runtime-types t) (runtime-types t₁)
runtime-types (⊢-reset₀ x t t₁ t₂) = ⊢-reset₀ (runtime-e x) (runtime-types t₂) (runtime-types t) (runtime-types t₁)

```

Small lemma about lifting context and that it keeps type safety.

```

open Typing
_#_ : Typing.Context → Typing.Context → Typing.Context
y # ∅ = y
y # (xs , x) = (y # xs) , x
e↑ : ∀ {Γ x A Γ'}
  → Γ ⊢ x :: A
  → Γ' # Γ ⊢ x :: A
e↑ : ∀ {Δ Θ Γ e T E Γ'}
  → Δ ; Θ ; Γ ⊢ e :: T / E
  → Δ ; Θ ; (Γ' # Γ) ⊢ e :: T / E

```

Substitution and proof of type preservation in substitution are defined at the same time.

```

module RExprSubstTyped where
  open RuntimeExpr
  ext : ∀ {Γ Γ'}
    → (∀ {A n} → Γ ⊢ n :: A → Σ[ m ∈ ℕ ] Γ' ⊢ m :: A)
    → (∀ {A B n} → (Γ , B) ⊢ n :: A → Σ[ m ∈ ℕ ] (Γ' , B) ⊢ m :: A)
  rename : ∀ {Γ Γ'}
    → (∀ {A n} → Γ ⊢ n :: A → Σ[ m ∈ ℕ ] Γ' ⊢ m :: A)
    → (∀ {Δ Θ A e E} → Δ ; Θ ; Γ ⊢ e :: A / E → Σ[ e' ∈ RExpr ] Δ ; Θ ; Γ' ⊢ e' :: A / E)
  exts : ∀ {Γ Γ' Δ Θ}
    → (∀ {n A E} → Γ ⊢ n :: A → Σ[ e ∈ RExpr ] Δ ; Θ ; Γ' ⊢ e :: A / E)
    → (∀ {n A B E} → Γ , B ⊢ n :: A → Σ[ e ∈ RExpr ] Δ ; Θ ; Γ' , B ⊢ e :: A / E)

```

```

subst :  $\forall \{ \Delta \Gamma \Gamma' \Theta \}$ 
   $\rightarrow (\forall \{ n A E \} \rightarrow \Gamma \ni n : A \rightarrow \Sigma[ e \in \text{RExp} ] \Delta ; \Theta ; \Gamma' \vdash e : A / E)$ 
   $\rightarrow (\forall \{ e A E \} \rightarrow \Delta ; \Theta ; \Gamma \vdash e : A / E \rightarrow \Sigma[ e \in \text{RExp} ] \Delta ; \Theta ; \Gamma' \vdash e : A / E)$ 

-[-] :  $\forall \{ \Delta \Theta \Gamma A B E1 \}$ 
   $\rightarrow (e \text{ e1} : \text{RExp})$ 
   $\rightarrow \{ te : \Delta ; \Theta ; \Gamma , A \vdash e : B / E1 \}$ 
   $\rightarrow \{ te1 : \forall \{ E \} \rightarrow \Delta ; \Theta ; \Gamma \vdash e1 : A / E \}$ 
   $\rightarrow \Sigma[ e' \in \text{RExp} ] \Delta ; \Theta ; \Gamma \vdash e' : B / E1$ 
-[-] { $\Delta$ }{ $\Theta$ }{ $\Gamma$ }{ $A$ }{ $B$ }{ $E1$ } e e1 {te}{te1} = subst { $\Delta$ }{ $\Gamma$  ,  $A$ }{ $\Gamma$ }  $\sigma$  te
  where
   $\sigma : (\forall \{ B n E \} \rightarrow \Gamma , A \ni n : B \rightarrow \Sigma[ e \in \text{RExp} ] \Delta ; \Theta ; \Gamma \vdash e : B / E)$ 
   $\sigma \{ E = E' \} Z = e1 ,, (te1 \{ E' \})$ 
   $\sigma \{ n = \text{suc } n \} (S x) = \text{var } n ,, \vdash \text{var } x$ 

```


Chapter 4

Progress

4.1 Values

Here we define values. Only abstractions, type abstractions and labels are considered values. Since values themselves don't perform any effects, they have `nil` effect. But rules for all of them have built-in weakening. We can use that to generalize their type and perform substitution where any effect is expected.

```
data Value : REpr → Set where
  vlam : ∀ { e } → Value (lam e)
  vLam : ∀ { k e } → Value (tLam k e)
  vlab : ∀ { n } → Value (label n)
gvalue : ∀ {Δ Θ Γ T E e} → (Value e) → (Δ ; Θ ; Γ ⊢ e :: T / E) → ∀ {F} → (Δ ; Θ ; Γ ⊢ e :: T / F)
gvalue vlam (⊢lam t) = ⊢lam t
gvalue vLam (⊢forall t) = ⊢forall t
gvalue vlab (⊢label x) = ⊢label x
gvalue {Δ} v (⊢weak x x1 x2) {F} = (⊢weak x ( <se-refl {Δ} {F} ) (gvalue v x2))
```

Frames would usually be named evaluation context, but here it's taken by typing context. Here frame represents parts between reset s, or reset and shift . Frame type is parametrized by $\Theta \Gamma$ - typing context outside of frame, T type of the hole. It's also indexed by Effects and Type of whole frame, Effects of the hole, typing context of the hole and amount of new' constructors - which says how many type binders are in the frame. Frames here are intrinsically typed, thus they also store type judgements of subexpressions.

```
data Frame (Θ : EContext) (Γ : Context) (T : Type) : Effects → Type → Effects → Set where
  fempty : ∀ {Eff}
    -----
    → Frame Θ Γ T Eff T Eff
  fapp1 : ∀ {A B Eff E} → Frame Θ Γ T Eff (A - Eff > B) E
    → (e : REpr) → { Δ ; Θ ; Γ ⊢ e :: A / Eff }
    -----
    → Frame Θ Γ T Eff B E
  fapp2 : ∀ {A B Eff E} → (e : REpr) → { v : Value e }
    → { Δ ; Θ ; Γ ⊢ e :: (A - Eff > B) / Eff }
    -----
    → Frame Θ Γ T Eff A E → Frame Θ Γ T Eff B E
  freset-label : ∀ {A E A' Eff C}
    → (e en : REpr)
    → Δ ; Θ ⊢ C :: e
    → Δ ; Θ ; Γ ⊢ e :: A' / (C :: Eff)
    → Δ ; Θ ; (Γ , A') ⊢ en :: A / Eff
    → Frame Θ Γ T nil (L C at A / Eff) E
```

```

-----
→ Frame Θ Γ T Eff A E
fshift-label : ∀ {A E A' E' C}
→ (e : RExpr)
→ ∅ ; Θ ⊢ C % e
→ ∅ ; Θ ; (Γ , A - E' > A' ) ⊢ e % A' / E'
→ Frame Θ Γ T nil (L C at A' / E') E
-----
→ Frame Θ Γ T (C :: nil) A E

```

Frame plugging and composition:

```

plug : ∀ {Θ Γ T Eff A E}
→ Frame Θ Γ T Eff A E
→ (e : RExpr) → ∅ ; Θ ; Γ ⊢ e % T / E
→ Σ[ res ∈ RExpr ] (∅ ; Θ ; Γ ⊢ res % A / Eff)
_of_ : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ Frame Θ Γ B Eff A Eff'
→ Frame Θ Γ C Eff' B Eff''
→ Frame Θ Γ C Eff A Eff''

```

We can prove how plugging and composition relate. Plugging expression into one frame and another results in same value and type as plugging expression into composition of two frames.

```

of-lemma : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ (f1 : Frame Θ Γ B Eff A Eff')
→ (f2 : Frame Θ Γ C Eff' B Eff'')
→ (e : RExpr) → (t : ∅ ; Θ ; Γ ⊢ e % C / Eff'')
→ plug (f1 of f2) e t
= ((λ x → plug f1 (Data.Product.proj₁ x) (Data.Product.proj₂ x))(plug f2 e t))

of-lemma fempty f2 e t = refl
of-lemma (fapp₁ f1 e₁) f2 e t rewrite of-lemma f1 f2 e t = refl
of-lemma (fapp₂ e₁ f1) f2 e t rewrite of-lemma f1 f2 e t = refl
of-lemma (freset-label ee en x x₁ x₂ f) f2 e t rewrite of-lemma f f2 e t = refl
of-lemma (fshift-label e₁ x x₁ f) f2 e t rewrite of-lemma f f2 e t = refl

```

Lifting frames into arbitrary context preserves types, same as with expressions.

```

↑f : forall { Θ A B Eff Eff' Γ' Γ }
→ Frame Θ Γ A Eff B Eff'
→ Frame Θ (Γ' * Γ) A Eff B Eff'

```

Metaframe stores whole evaluation context, it's split into frames separated by resets. Type parameters and indices work the same as in frame. Unlike in frame, metaframe now stores resets, so lists of effects inside and outside of frame may differ. That means their difference represents list of effects handled by the frame. This observation can be used to prove that for well typed expression (in empty typing context, and with condition that same labels have the same type) that decomposes into metaframe and `shift₀` expression, and metaframe should handle effect of the `shift`. Also this metaframe decomposes into two metaframes separated by `reset₀` which has same label as mentioned shift.

```

data Metaframe (Θ : EContext) (Γ : Context) (T : Type) (Eff : Effects)
: Type → Effects → Set where
mfempty : Metaframe Θ Γ T Eff T Eff
mfreset : ∀ { A B C Eff' }
→ (l : Label)
→ ∅ ; Θ ⊢ C % e
→ ∅ ; Θ ; Γ ⊢ label l % (L C at B / Eff) / nil
→ (e : RExpr) → (∅ ; Θ ; Γ , A ⊢ e % B / Eff)
→ Metaframe Θ Γ T (C :: Eff) A Eff'

```

```

-----
→ Metaframe Θ Γ T Eff B Eff'
mframe : ∀ {A Eff' B Eff''}
→ Frame Θ      Γ A Eff B Eff'
→ Metaframe Θ Γ T Eff' A Eff''
-----
→ Metaframe Θ Γ T Eff B Eff''

```

Metaframes, same as frames, can be lifted into arbitrary contexts, plugged and composed. They can also be composed with simple frames.

```

tm : forall {Θ A B Eff Eff' Γ' Γ}
→ Metaframe Θ Γ      A Eff B Eff'
→ Metaframe Θ (Γ' + Γ) A Eff B Eff'
mplug : ∀ {Θ Γ T Eff A E}
→ Metaframe Θ Γ T Eff A E
→ (e : REpr) → ∅ ; Θ ; Γ ⊢ e :: T / E
→ Σ[ res ∈ REpr ] (∅ ; Θ ; Γ ⊢ res :: A / Eff)
_om_ : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ Metaframe Θ Γ B Eff A Eff'
→ Metaframe Θ Γ C Eff' B Eff''
→ Metaframe Θ Γ C Eff A Eff''
_fom_ : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ Frame Θ Γ B Eff A Eff'
→ Metaframe Θ Γ C Eff' B Eff''
→ Metaframe Θ Γ C Eff A Eff''

```

4.2 Reduction

Since labels need to be allocated, reduction relation is defined in terms of expression and state. State itself is just next label to be allocated. As frames are intrinsically typed, we need to provide judgment representing well-typedness of expressions.

```

infix 2 _→_
State = EContext
data _→_ : REpr × State → REpr × State → Set where
{-
→new : ∀ {e Δ Θ E T A1 E1}
→ {te : (Δ , Kind.E) ; Θ ; (∅ , L ttv zero at A1 / E1) ⊢ e :: E / T}
→ new e , ' Θ → REprSubstTyped._[_] e (label (Θ .proj₁)) {te = te} {te1 = {!!}} .proj₁ , ' Θ --(Data.Vec._++_ Θ ( Data.Vec
-}
β-lam-app : ∀ {e V Θ Γ A B E}
→ {te : ∅ ; Θ ; (Γ , A) ⊢ e :: B / E}
→ {tv : ∅ ; Θ ; Γ ⊢ V :: A / E}
→ Value (lam e)
→ (v : Value V)
→ app (lam e) V , ' Θ → (proj₁ (REprSubstTyped._[_] e V {te = te} {te1 = (gvalue v tv)})) , ' Θ

β-tlam-tapp : ∀ {k e T Θ}
→ Value (tlam k e)
→ tapp (tlam k e) T , ' Θ → e REprSubst.e[t T] , ' Θ

reset₀-vl : ∀ {V e' en Θ Γ A B E}
→ ( v : Value V)
→ {tv : ∅ ; Θ ; Γ ⊢ V :: A / E}
→ {ten : ∅ ; Θ ; (Γ , A) ⊢ en :: B / E}
→ reset₀ V en e' , ' Θ → (REprSubstTyped._[_] en V {te = ten} {te1 = (gvalue v tv)} .proj₁) , ' Θ

{-
reset₀-k : ∀ {es en e' e s n Θ A T Eff Eff' B A' E' ls lr}
→ { f : Metaframe Θ ∅ T Eff A Eff' }
→ ∀ {cont-type}
→ Value (e')
--shift
→ {ts : ∅ ; Θ ; ∅ ⊢ (shift₀ e' es) :: T / Eff'}
→ {tes : ∅ ; Θ ; (∅ , T - Eff' > B) ⊢ es :: B / Eff' }
→ {tls : ∅ ; Θ ; ∅ ⊢ e' :: (L ttv ls at B / Eff') / nil }
→ {tlvs : ∅ ; Θ ⊢ ttv lr :: e }
--reset
→ {te : Δ ; ∅ ⊢ e :: A' / (ttv lr :: E') }

```

```

→ {ten : Δ ; ∅ , A' ⊢ en s T / Eff }
→ {tlr : Δ ; ∅ ⊢ e' s (L ttv lr at T / Eff) / nil }
→ {tlvr : Δ ⊢ ttv lr se }
→ (proj1 (mplug f (shift0 e' es) ts)) ≡ e
→ reset0 e en e' , ' s
→ REExprSubstTyped...[_] es
  (lam (reset0 (proj1 (mplug (↑m {Γ' = ∅ , T} f) (var 0) (←var Z))) en e'))
  {te = tes} {te1 = gvalue {E = Eff} vlam cont-type}
  .proj1 , ' s
-}

```

Since simple reduction above is defined directly on redexes, we introduce $\rightarrow$ that represents reduction within metaframe. As we are only considering whole typed expressions, in place of Γ we use empty context.

```

{-
infix 2 _→_
data _→_ : REExpr × State → REExpr × State → Set where
  →frame : ∀ {e1 e1' e2 e2' s s' n Δ Δ' A T Eff Eff' t1 t2} → (f : Metaframe Δ ∅ T Eff A Eff' Δ' n)
    → e1' , ' s → e2' , ' s'
    → Data.Product.proj1 (mplug f e1' t1) ≡ e1
    → Data.Product.proj1 (mplug f e2' t2) ≡ e2
    → (e1 , ' s) → (e2 , ' s')
-}

```

4.3 Progress

```

{-
data Decompose : State → (e : REExpr) → Set where
  de-simpl-redex : ∀ {e e2 s s' n Δ Δ' A T Eff Eff'}
    → (f : Metaframe Δ ∅ T Eff A Eff' Δ' n)
    → (e , ' s) → (e2 , ' s')
    → (t : Δ ; ∅ ⊢ e s A / Eff)
    → Decompose s e
  de-shift : ∀ {s Δ Δ' T Eff A n Eff' es es' e l t}
    → (f : Metaframe Δ ∅ T Eff A Eff' Δ' n)
    → shift0 (label l) es' ≡ es
    → Data.Product.proj1 (mplug f es t) ≡ e
    → (t : Δ ; ∅ ⊢ e s A / Eff)
    → (ts : Δ ; ∅ ⊢ es s T / Eff')
    → Decompose s e
  de-val : ∀ {Δ Eff A s e} → { t : Δ ; ∅ ⊢ e s A / Eff }
    → Value e
    → Decompose s e
data Progress : State → REExpr → Set where
  done : ∀ {e s} → Value e → Progress s e
  step : ∀ {e1 s1 e2 s2}
    → ( e1 , ' s1 ) → ( e2 , ' s2 )
    → Progress s1 e1
-}

```

Proof of progress would have a type of $\text{progress} : \forall \{A \Delta \text{Eff}s\} \rightarrow (s : \text{State}) \rightarrow (e : \text{REExpr}) \rightarrow (t : \Delta ; \emptyset \vdash e s A / \text{Eff}s) \rightarrow \text{Progress } s e$. In such proof We would use auxiliary struct **Decompose** which builder would walk down well typed expression recursively until it has reached either value, simple reduction (app, tapp, new), or shift, and return it with surrounding metaframe.

In case of shift, such metaframe by construction should have effect handler that has same effect as shift. So we can construct **rest₀-k** and surrounding metaframe. Other cases would either be immediate value, or simple reduction in context.

4.3.1 Preservation

Current definition of reduction relation would yield proof of preservation immediately if progress was given since, frames and expressions are well typed, and reduction relation requires proofs to plug into metaframes or substitution.

Chapter 5

Discussion/Closing thoughts

In this paper we presented language with labeled delimited effect handlers together with partial formalisation of such language. We defined 3 term languages, typing judgements for 2 of them, translations between those languages and that typing is preserved. We defined intrinsically typed frames and metaframes that allow us to define reduction relation, how it preserves types. Since they are typed we statically know what effects are handled by frame. We also defined draft of abstract machine and how it could evaluate this language. This formalisation could be expanded and finished to fully prove correctness of the language and translations.

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